

PROJECT DOCUMENTATION

Script-Controlled ACL – Restrict Record Access Based on Field Value

INTRODUCTION:

**PROJECT TITLE: Script-Controlled
ACL – Restrict Record Access Based on
Field Value**

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Script-Controlled ACL – Restrict

Record Access Based on Field Value

Description:

This lab demonstrates how to implement a script-controlled Access Control List (ACL) to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.

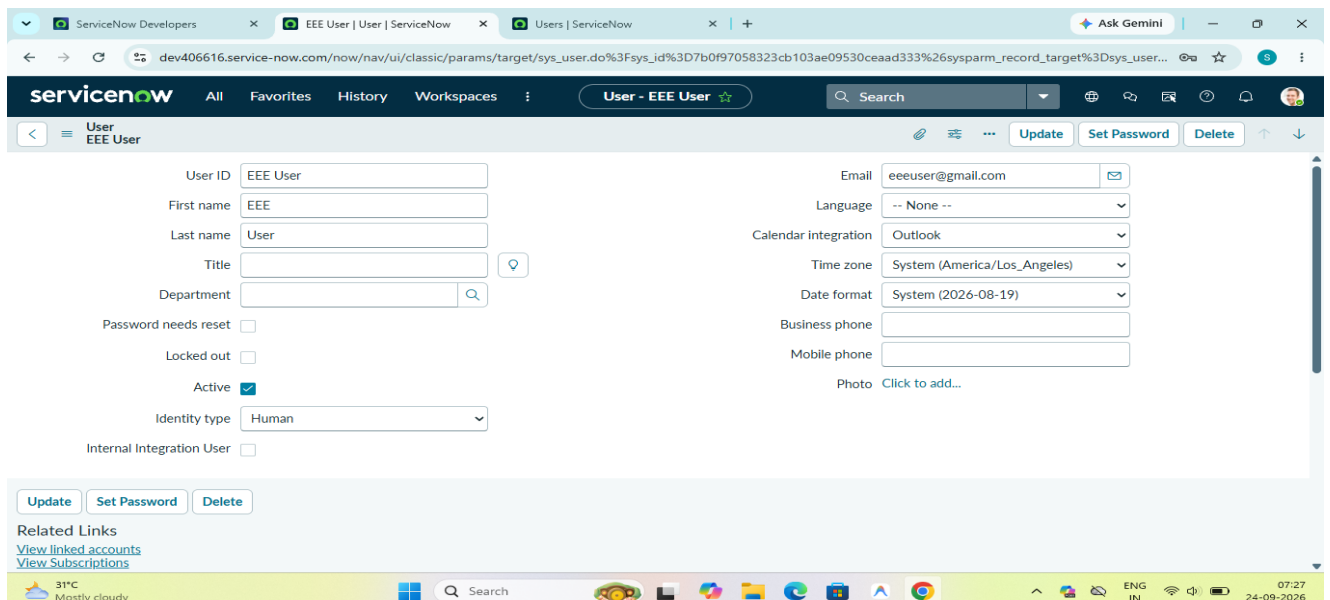
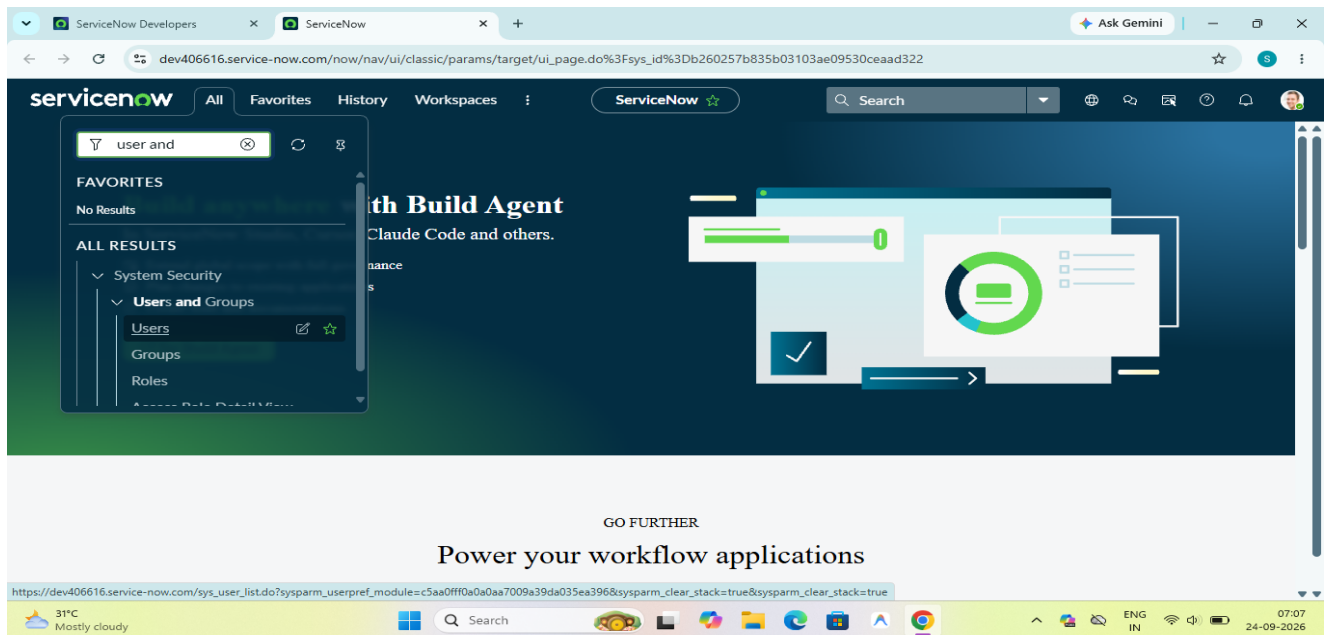
Timer:

3 hrs

Milestone-1 Creation of Users and Roles

User Creation:

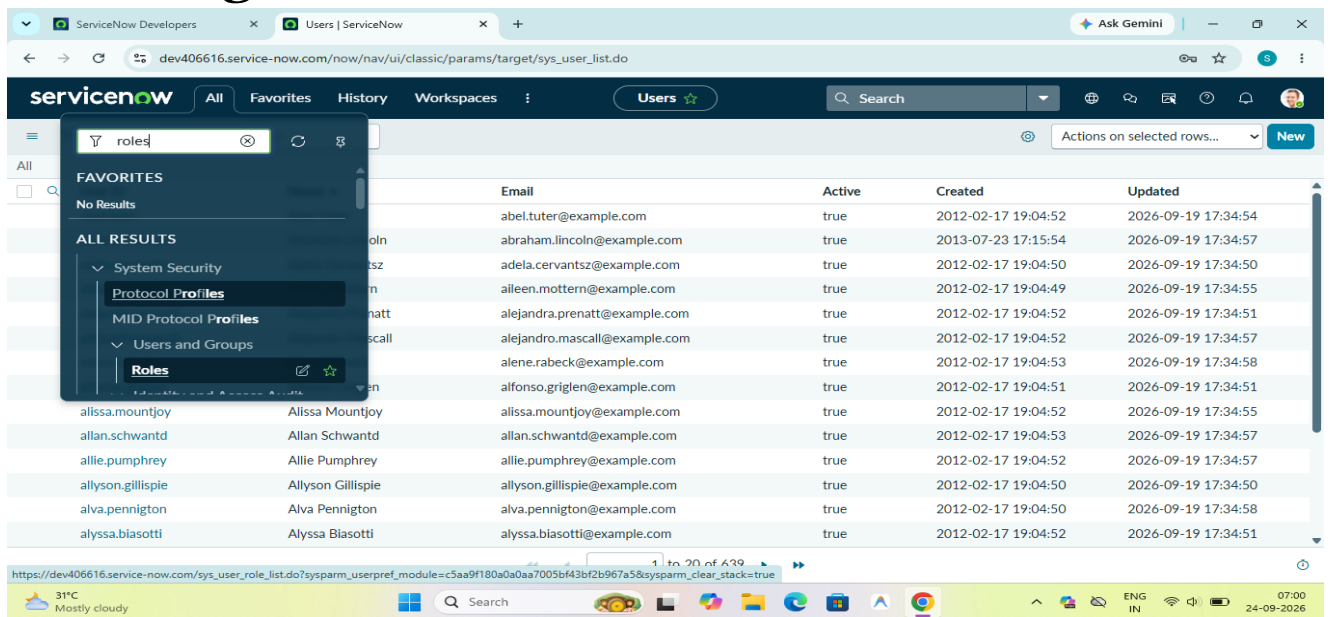
- 1.Log in to the ServiceNow instance with admin access.**
- 2.Create a test user:**
 - **Navigate to User Administration--> Users**
 - **Click New**
 - **User ID: EEE User**
 - **First Name: EEE**
 - **Last Name: User**
 - **Email: eeeuser@gmail.com**
 - **Save the record.**



Role Creation

- **Navigate to Administration → Roles**
- **Click New**
- **Name: Enter role name as bb1**
- **Description: Brief role purpose**

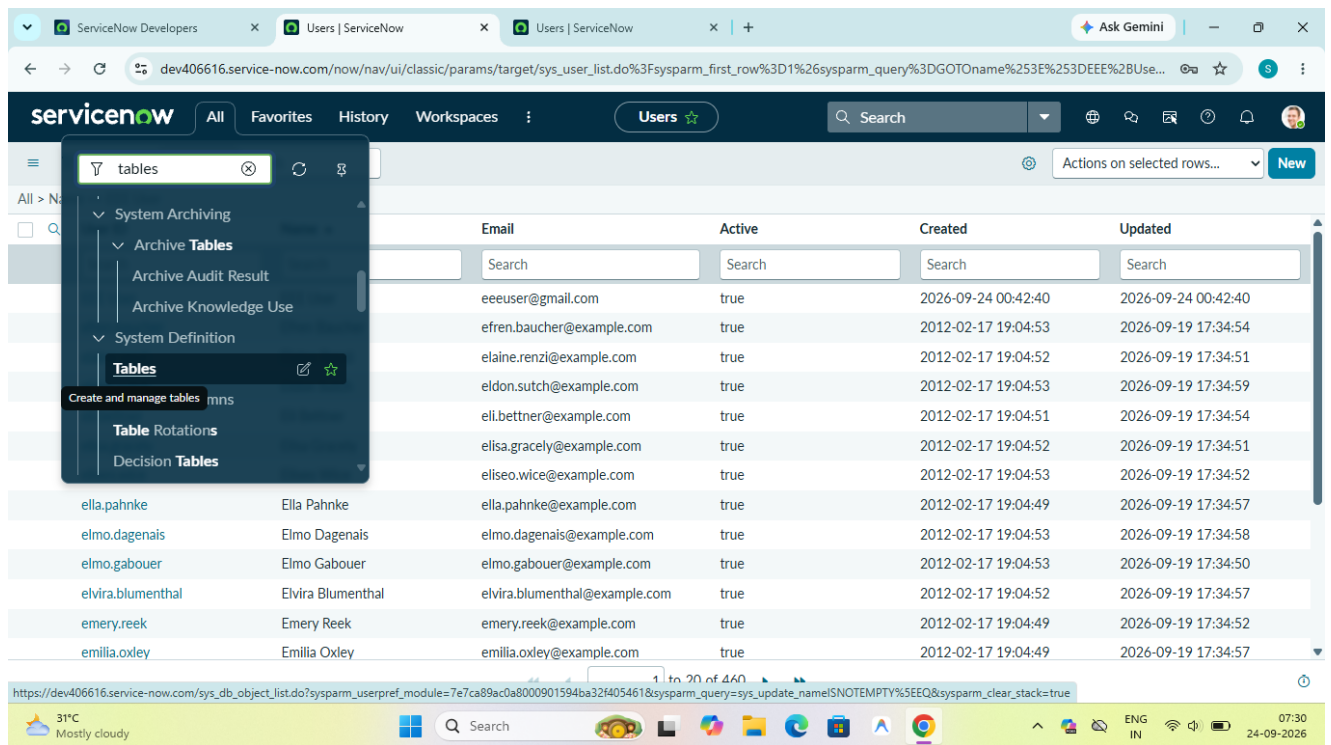
- Click Submit
- Create other roles also like bb2,bb3,bb4.(follow the same procedure)
- Assign all the roles to the EEE User



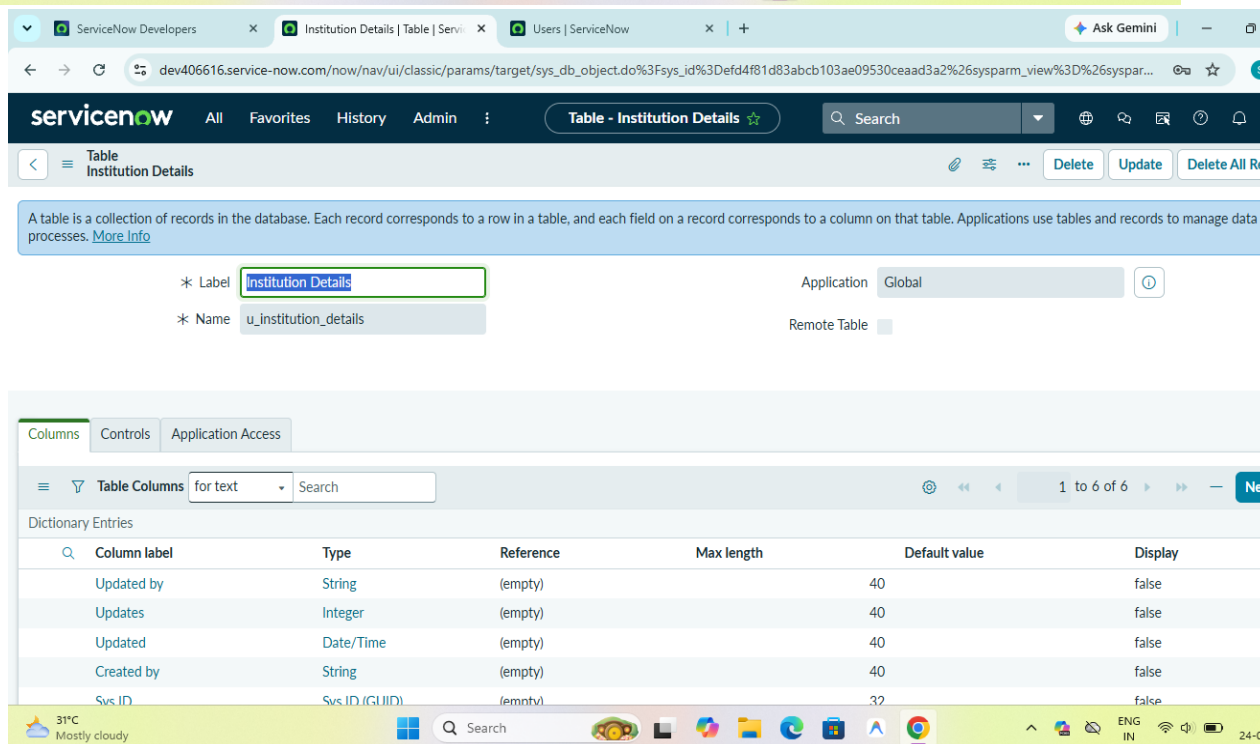
Tables Creation

1. In the Application Navigator, search for System Definition → Tables.
2. Click New and create a custom table with the following details:
 - Label: Institution Details
 - Name: u_institution_details

• Extends for : false



The screenshot shows the ServiceNow 'Users' list page. A dropdown menu is open from the 'Tables' icon in the left sidebar, showing options like 'System Archiving', 'Archive Tables', 'System Definition', and 'Tables'. The 'Users' table is visible in the background, showing columns for Email, Active, Created, and Updated. The 'Email' column contains various email addresses, and the 'Active' column is set to 'true' for all users.



The screenshot shows the 'Table - Institution Details' configuration page in ServiceNow. The 'Label' field is set to 'Institution Details' and the 'Name' field is set to 'u_institution_details'. The 'Application' is set to 'Global'. Below the configuration fields, there is a 'Columns' tab with a table of dictionary entries.

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Updated	Date/Time	(empty)		40	false
Created by	String	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false

3. Click Submit.

4. Add the following fields to the u_institution_details table:

- **Student Roll Number– Auto Number**
- **Student Name – Reference- User**
- **Faculty Name – Reference- User**
- **Branch – Choice (ECE, EEE, CSE)**
- **Email-String**
- **PhoneNumber–String**
- **Description– Multi String**

5. Save the Table

6. Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).

The screenshot shows the ServiceNow 'Institution Details' page. It features a table with the following data:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Andrew Och		Amos Linnan	STU300
ECE			Abraham Lincoln		Abel Tuter	STU100
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

Below the table, there is a pagination bar showing '1 to 3 of 3' and a status bar at the bottom indicating '31°C Mostly cloudy' and the date '24-09-2026'.

Creation of Access Control Lists(Read ACL)

1. In the Application Navigator, search for System Security → Access Control (ACL).
2. Elevate “Security_admin” role
3. Click New to create a new ACL.
4. Enter the following details:
 - Type: record
 - Operation: read
 - Name:

u_institution_details .

Active: true .Advanced: true

5. In the Requires role related list, add: .

bb1 (custom role)

6. Add Data condition as Branch is EEE

7. In the Script field, add the following script:

```
(function () {
```

```
    // Allow admin users full access
```

```
    if (gs.hasRole('admin')) {
```

```
return true;
```

```
}
```

```
    // Allow only EEE branch users to see EEE records
```

```
    if (gs.hasRole('bb1')){
```

```
return true;
```

```
}
```

```
    // Deny access for all others    return
```

```
false;
```

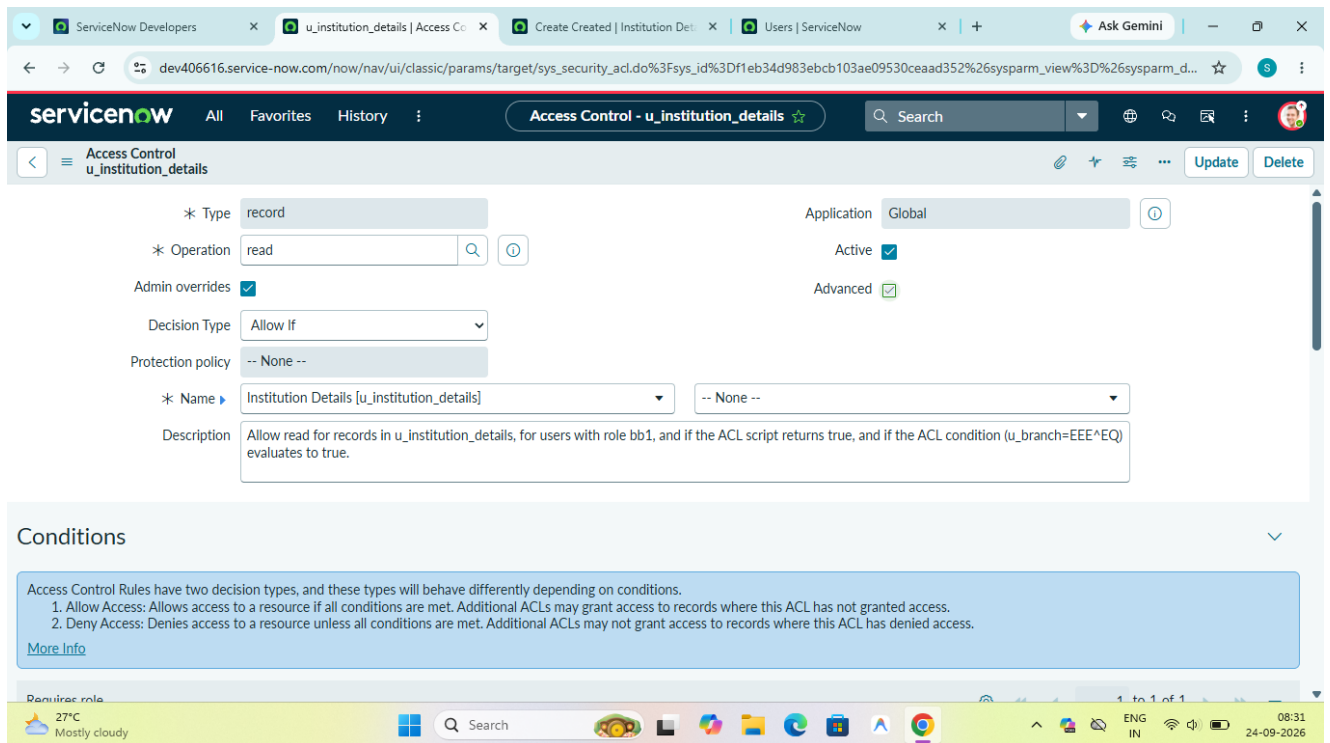
```
})();
```

8. Click Submit to save the ACL.

The image consists of two screenshots of the ServiceNow web interface, illustrating the steps to save an ACL.

Top Screenshot: The user is logged in as a System Administrator. The top navigation bar includes 'ServiceNow' and a search bar. The main content area features a 'Build anywhere with Build Agent' banner. On the right, a user profile dropdown menu is open, showing options like 'Profile', 'Preferences', 'Keyboard shortcuts', 'Impersonate user', 'Elevate role', 'Printer friendly version', and 'Log out'.

Bottom Screenshot: The user has navigated to the 'Access Control (ACL)' configuration page. The left sidebar shows a search bar with 'Access d' entered, and a list of results under 'ALL RESULTS' including 'Access Control (ACL)', 'Machine Identity Access Control', 'IP Address Access Control', and 'Processor Access Policies'. The main content area shows the 'Build Agent' banner. The bottom status bar displays the URL: `https://dev406616.service-now.com/sys_security_acl_list.do?sysparm_userpref_module=4ec206740a0aa700b850e0337b2f00&sysparm_clear_stack=true`.



9. Impersonate EEE users and open the Student Records list.

10. Copy the table name `u_institution_details.list/LIST` and paste in the All menu.

11. Impersonate Admin and open the Student Records list.

Verify :

- User with bb1 role can view only EEE branch records

The screenshot shows the ServiceNow 'Institution Details' page. At the top, there's a navigation bar with 'All', 'Favorites', and 'History' tabs. Below it, a search bar is visible. The main content area displays a table with the following columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. A single record is shown with the following data:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Alyson Gillisple		Alejandra Prenatt	STU200

At the bottom of the screenshot, a Windows taskbar is visible with the date 24-09-2026 and time 08:45.

- User without the role cannot view any records
- Admin user can view all records regardless of branch

Creation of Access Control Lists(Create ACL)

1. Click New to create a new ACL(Create)
2. Enter the following details:
 - o Type: record
 - o Operation: create
 - o Name: u_institution_details
 - o Active: true
3. In the Requires role related list, add: o bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

The screenshot displays the ServiceNow 'Access Control - New Record' form. The top navigation bar shows 'ServiceNow' and 'Access Control - New Record'. The form fields are as follows:

- * Type: record
- * Operation: create
- Application: Global
- Active: ☒
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: (empty text area)

A warning message is displayed at the top of the form: "Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL."

Below the form is a 'Conditions' section with the following text:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

Verify :

- User with bb1 &bb2 roles can view only
EEE branch records b New Button

Creation of Access Control Lists(Write ACL)

12. Click New to create a new ACL(Write)

13. Enter the following details:

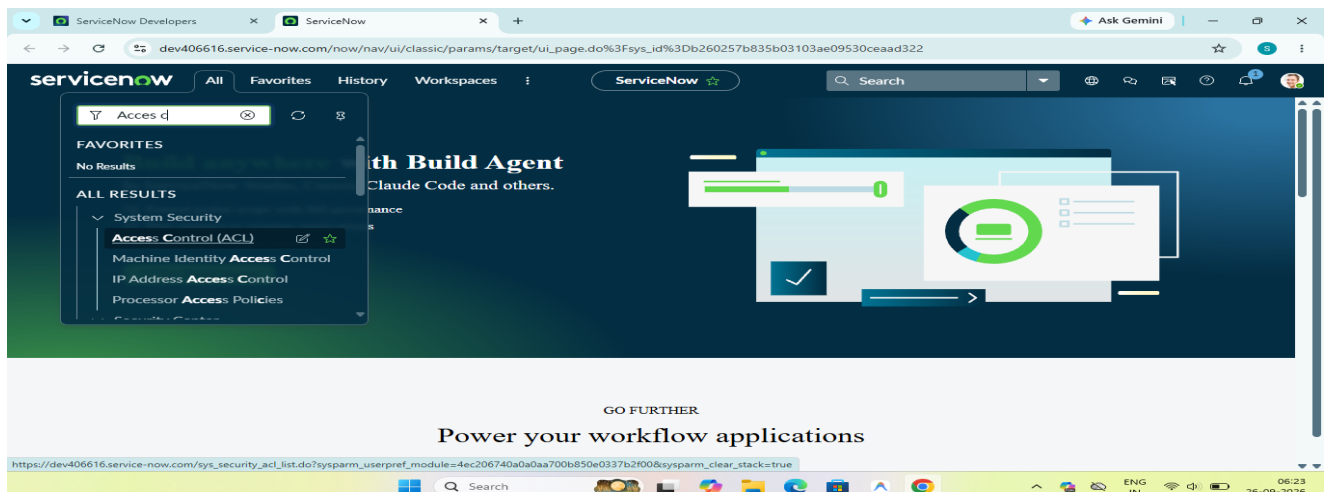
- Type: record
- Operation: Write
- Name: u_institution_details
- Active: true

14. In the Requires role related list, add:

- bb3 (custom role)

15. No need to add any Data condition

16. Save the ACL Record



Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: write

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Application: Global

Active: ☒

Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

Verify :

- User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button

Creation of Access Control Lists(Delete ACL)

17. Click New to create a new ACL(Delete)

18. Enter the following details:

- o Type: recor
- o Operation: Delete
- o Name: *uinstitution*
- o Active: true

19. In the Requires role related list, add: o bb4
(custom role)

20. No need to add any Data condition

21. Save the ACL Record

The screenshot displays the ServiceNow web interface for configuring an Access Control List (ACL) record. The browser address bar shows the URL: `dev406616.service-now.com/now/nav/ui/classic/params/target/ui_page.do%3Fsys_id%3D%3Db260257b835b03103ae09530cead322`.

The top navigation bar includes the ServiceNow logo, tabs for 'All', 'Favorites', 'History', and 'Workspaces', a search bar, and a user profile icon. A dropdown menu is open, showing 'Access Control (ACL)' as the selected option under the 'System Security' category.

The main content area is titled 'GO FURTHER Power your workflow applications'. Below this, the browser address bar shows the URL: `https://dev406616.service-now.com/sys_security_acl_list.do?sysparm_userpref_module=4ec206740a0aa700b850e0337b2f00&sysparm_clear_stack=true`.

The 'Access Control - New Record' form is displayed, featuring a warning message: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The form includes the following fields:

- * Type: record
- * Operation: delete
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: (empty text area)
- Application: Global
- Active: ☒
- Advanced: ☐

The 'Conditions' section is expanded, showing a blue information box with the following text:

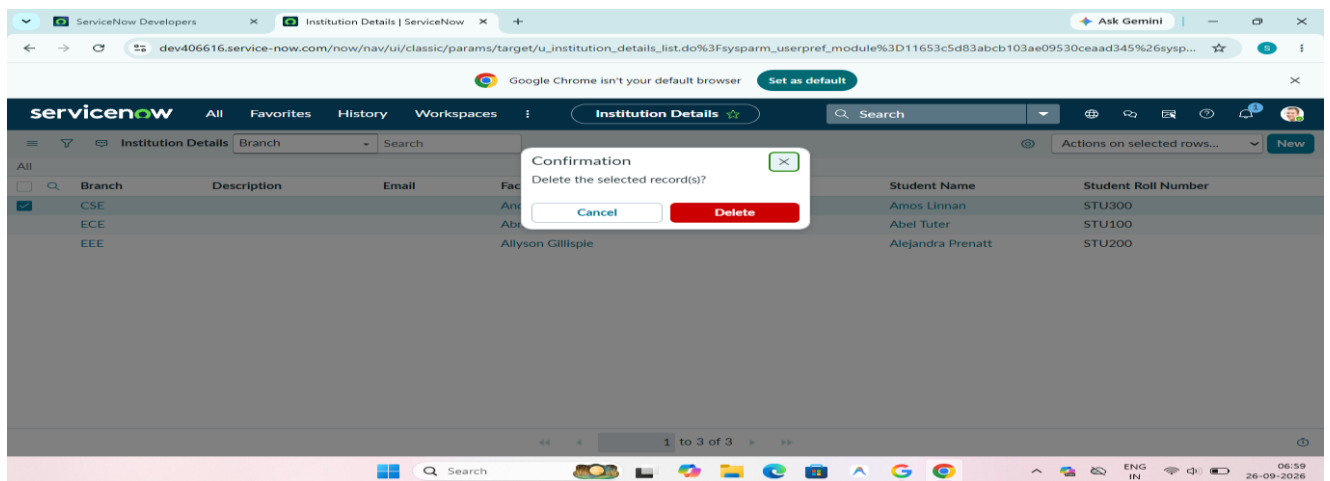
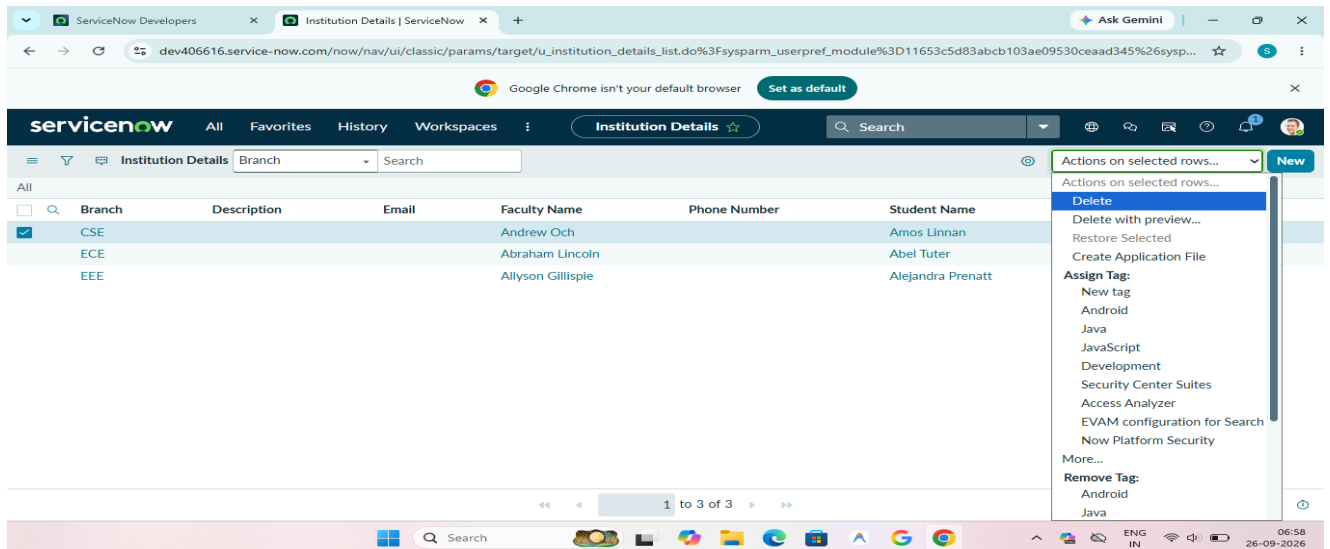
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1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

The bottom of the screen shows a Windows taskbar with the date and time: 08:40 24-09-2026.

Verify :

- Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the EEE branch records.



Outcome :

By completing this lab, learners gain a comprehensive understanding of how scriptcontrolled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how

user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.